Proofs!

COMP0147

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Proof Techniques

Introduction

- A **proof** is a mathematical demonstration of certainty about a statement.
- The level of detail in a proof may depend on the type of reader it is intended for, taking into account factors such as:
  - the reader's knowledge of the subject;
  - the reader's maturity;
  - the level of rigor desired.
- In this lecture, we will focus on proofs using the mathematical rigor expected of a professional at the undergraduate level in the exact sciences.
- Proofs are important in several areas of Computer Science:

1. program correctness;
2. algorithm complexity analysis;
3. system security properties;
4. ...

Terminology

- An **axiom (or postulate)** is a statement assumed to be true without the need for proof; axioms are considered “”self-evident truths””.
- A **theorem** is a statement that can be shown to be true. A theorem is a result considered interesting in itself.
- A **proposition** is also a statement that can be shown to be true, but is considered a theorem of “”lesser interest””.
- A **lemma** is an auxiliary statement to be proved, usually to break a large theorem proof into smaller pieces.
- A **proof (or demonstration)** is an argument that shows a statement (theorem, proposition, or lemma) follows from a set of premises.
- A **corollary** is a statement easily derived from an already proved theorem. Corollaries are immediate consequences of other results.
- A **conjecture** is a well-founded assumption, but (so far) without proof. Once proved, a conjecture becomes a theorem or a proposition.

Evidence versus Proof

- Let the formula $p(n) = n^2 + n + 41$.

  Conjecture: $\forall n \in \mathbb{N}.p(n)$ is prime.

  Evidence that the conjecture is correct:

  Testing values $n = 0, 1, \ldots, 39$ the statement is always true, i.e., $p(n)$ is prime for $0 \leq n \leq 39$.

n	0	1	2	3	...	20	...	39
$p(n)$	41	43	47	53	...	461	...	1601

  This cannot be a coincidence! The hypothesis must be true!

  But it is not: $p(40) = 1681 = 41 \cdot 41$, which is not prime!

  Thus, the conjecture is false.

- Evidence is not the same as proof!

Evidence versus Proof

- In 1769, Euler (1707–1783) conjectured that

$$a^4 + b^4 + c^4 = d^4$$

has no solution in the set of positive integers.

For more than two centuries, no one could find values of a, b, c and d that satisfied the equation.

The failure of all mathematicians involved was evidence that the conjecture might be true.

218 years later, in 1987, Noam Elkies provided a counterexample, showing that

$$95\,800^4 + 217\,519^4 + 414\,560^4 = 422\,481^4.$$

Thus, this conjecture is also false.

- Absence of proof is not proof of absence!

Proof Techniques

- Constructing a proof is an art.

  Each case is different: there is no "one-size-fits-all recipe" for constructing proofs for all statements.

- There are, however, techniques that are useful to prove a large number of statements. In this lecture, we will analyse the following:

  1. direct proof;
  2. proof by contraposition;
  3. proof by contradiction (or proof by reduction to absurdity).
  4. proof by counterexample;
  5. proof by exhaustion and case analysis;

- Other proof techniques (e.g., proof by mathematical induction) will be seen later in the course.

How to Write a Proof

- Clearly state the statement you want to prove. (It is common to precede the statement with the one of the words **Theorem**, **Proposition**, **Lemma** or **Conjecture**.)
- Clearly delimit the scope of the proof.

  Indicate the start of the proof with the word **Proof.**

  Indicate the end of the proof with a marker. One can use:

  - a little square □
  - the abbreviation **qed** (from Latin "quod erat demonstrandum")
- Write the proof in such a way that it is self-contained. Use natural language clearly, employing complete and well-structured sentences. Mathematical formulas, equations, etc., can be used when necessary.

How to Write a Proof

- Identify each variable used in the proof along with its type. Examples:
  1. Let x be a real number greater than 2.
  2. Suppose m and n are integers with no common divisors.
- Important:

  The main goal of a proof is to convince the reader that the result (theorem, proposition, lemma) is true.

  It is not enough that you yourself are convinced!

  Make sure you are concise but clear.

Direct Proof

- General form:

  1. Express the statement to be proved in the form:

     $$\forall x \in D.(P(x) \rightarrow Q(x))$$

     This step is sometimes done mentally.

  2. Start the proof by assuming that x is a specific element of the domain D, but chosen arbitrarily, for which the hypothesis $P(x)$ is true.

     We usually abbreviate this step by saying "Let $x \in D$, and assume that $P(x)$ holds." or "Let $x \in D$ be such that $P(x)$".

  3. Show that the conclusion $Q(x)$ is true using definitions, previous results, and the rules of logical inference.

- Important: Since $x \in D$ is chosen arbitrarily,
  - it does not depend on any special assumption about x, and
  - therefore, it can be generalized to all elements of D.

Direct Proof

Example 1: Show that if n is an odd integer, then n^2 is odd.

(**Note:** An integer n is even iff there exists an integer k such that $n = 2k$. An integer n is odd iff there exists an integer k such that $n = 2k + 1$.)

Proof. We want to show that $\forall n.(P(n) \rightarrow Q(n))$, where $P(n)$ is the predicate "n is an odd integer", and $Q(n)$ is the predicate "n^2 is odd".

To produce a direct proof of this statement, we assume that the hypothesis of the implication, $P(n)$, is true, i.e., that n is an odd integer. Then, by the definition of an odd number, there exists an integer k such that $n = 2k + 1$.

We want to show that the conclusion of the implication, $Q(n)$, is true, i.e., that n^2 is also odd. For this, we can calculate

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Thus, by the definition of an odd number, n^2 is also odd.

□

Direct Proof

Example 2: Show that if m and n are perfect squares, then mn is a perfect square.

(**Note:** An integer a is a perfect square if there exists an integer b such that $a = b^2$.)

Proof. To prove this proposition, assume that m and n are perfect squares. By the definition of a perfect square, there exist integers s and t such that $m = s^2$ and $n = t^2$.

The goal of the proof is to show that mn will be a perfect square when m and n are. To see this, we can calculate

$$mn \quad = \quad s^2t^2 \quad = \quad (st)^2.$$

Clearly, st is also an integer, so mn satisfies the definition of a perfect square (since $mn = (st)^2$), and the conclusion of the implication is true.

Thus, we conclude the proof that the statement is true. □

□

Direct Proof

Example 3: Show that the sum of two rational numbers is a rational number.

(**Note:** A number n is rational if there exist integers p and q, with $q \neq 0$, such that $n = p/q$.)

Proof. To give a direct proof of this statement, assume that r and s are rational. By the definition of a rational number, there exist integers p, q, t, u with $q, u \neq 0$, such that $r = p/q$ and $s = t/u$.

Thus

$$r + s \quad = \quad \frac{p}{q} + \frac{t}{u} \quad = \quad \frac{pu + qt}{qu}.$$

Note that, by hypothesis, q and u are nonzero, so $qu \neq 0$.

Consequently, $r + s$ can be expressed as the ratio of two integers ($pu + qt$ and qu, with $qu \neq 0$), and therefore $r + s$ satisfies the definition of a rational number.

Thus the statement is true. □

Proof by Contraposition

- General form:
  1. Express the statement to be proved in the form:
  $$\forall x \in D.(P(x) \rightarrow Q(x))$$
  2. Find the contrapositive of the statement:
  $$\forall x \in D.(\neg Q(x) \rightarrow \neg P(x))$$
  3. Start the proof by assuming that x is a specific element of the domain D, but chosen arbitrarily, for which the conclusion $Q(x)$ is false.
  4. Show that the hypothesis $P(x)$ is false using definitions, previous results, and rules of logical inference.
- Important: Since $x \in D$ is chosen arbitrarily,
  - it does not depend on any special assumption about x, and
  - therefore, it can be generalized to all elements of D.

Proof by Contraposition

Example 4: Show that if n is an integer and $3n + 2$ is odd, then n is odd.

Proof. We want to show that $\forall n \in \mathbb{N}.(P(n) \to Q(n))$, where $P(n)$ is "$3n + 2$ is odd", and $Q(n)$ is "n is odd".

To produce a proof by contraposition, we show that $\forall n \in \mathbb{N}.(\neg Q(n) \to \neg P(n))$. That is, we show that if an integer n is not odd, then $3n + 2$ is not odd.

If n is not odd, it is even, and by the definition of even numbers, $n = 2k$ for some $k \in \mathbb{N}$. Then we can derive

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1), \end{aligned}$$

from which we conclude that $3n + 2$ satisfies the definition of an even number.

Since we have shown that whenever the conclusion of the implication is false, the hypothesis is also false, the proof by contraposition is successfully completed. □

Proof by Contraposition

Example 5: Show that if $n = ab$ where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

Proof. To produce a proof by contraposition, we show that whenever the conclusion of the implication is false, the hypothesis is also false.

The conclusion of the implication can be written as $(a \leq \sqrt{n}) \vee (b \leq \sqrt{n})$.

Thus, by De Morgan's law, the negation of the conclusion is

$$\begin{aligned} \neg((a \leq \sqrt{n}) \vee (b \leq \sqrt{n})) &\equiv \neg(a \leq \sqrt{n}) \wedge \neg(b \leq \sqrt{n}) \\ &\equiv (a > \sqrt{n}) \wedge (b > \sqrt{n}). \end{aligned}$$

The hypothesis of the implication can be written as $n = ab$, and its negation is $n \neq ab$.

Proof by Contraposition

We want to show that if $(a > \sqrt{n}) \wedge (b > \sqrt{n})$, then $n \neq ab$.

For this, note that if $(a > \sqrt{n}) \wedge (b > \sqrt{n})$, we can derive

$$ab \quad > \quad \sqrt{n}\sqrt{n} \quad = \quad n,$$

from which we conclude that $ab > n$, and therefore $ab \neq n$.

Since we have shown that whenever the conclusion of the implication is false, the hypothesis is also false, the proof by contraposition is successfully completed.

□

Proof by Contradiction

- Proof by contradiction is based on the fact that if starting from a premise P, and applying inference rules correctly, we arrive at a false conclusion, then the premise P must necessarily be false.
- Equivalently, if assuming the negation $\neg P$ of a statement leads to an absurdity (contradiction), then the statement p must necessarily be true.
- General form:
  1. To prove that the statement P is true, assume that its negation $\neg P$ is true.
  2. Show that $\neg P$ leads to a contradiction, that is, that

$$\neg P \rightarrow F.$$

Proof by Contradiction

Example 6: Show that in a group of 22 days, at least 4 days fall on the same day of the week.

Proof. Let P be the proposition "In a group of 22 days, at least 4 days fall on the same day of the week".

Assume that $\neg P$ is true, that is, that "In a group of 22 days, at most 3 days fall on the same day of the week". In this case, at most 21 days can be chosen to be part of a group (since there are only 7 days in a week). But this contradicts the premise that the group has 22 days.

In other words, if R is the proposition "22 days are chosen", we would have $\neg P \rightarrow (R \wedge \neg R)$, that is, $\neg P \rightarrow F$.

Thus, $\neg P$ cannot be true, and therefore P is true. $\square$

Proof by Contradiction

Example 7: Show that if $3n + 2$ is odd, then n is odd.

Proof. We want to show the proposition "if $3n + 2$ is odd, then n is odd". We can write this proposition as $P \to Q$.

To prove by contradiction, assume that $P \to Q$ is false. This means we are assuming $P \wedge \neg Q$, that is, "$3n + 2$ is odd and n is not odd".

But if n is not odd, it is even, and there exists an integer k such that $n = 2k$. Then we can derive

$$3n + 2 \quad = \quad 3(2k) + 2 \quad = \quad 6k + 2 \quad = \quad 2(3k + 1),$$

which implies that $3n + 2$ is even. But this means we have exactly concluded that P is false, which contradicts the assumption that P is true.

Thus, it is impossible to have $p\P \wedge \neg Q$ without reaching a contradiction, and therefore if $3n + 2$ is odd, then n is odd. □

Proof by Contradiction

Example 8: $\sqrt{2}$ is irrational.

Proof. Assume the contrary of what we want to prove, that is, that $\sqrt{2}$ is rational.

In this case, there exist $p, q \in \mathbb{Z}$, with $\gcd(p, q) = 1$, such that $\sqrt{2} = p/q$. Squaring both sides, we obtain $2 = p^2/q^2$, that is, $p^2 = 2q^2$. Note that $2q^2$ is even, so by the equality above p^2 must also be even. This implies that p must be even.

Now, since p is even, there exists some $s \in \mathbb{Z}$ such that $p = 2s$. This implies that $2q^2 = p^2 = (2s)^2 = 4s^2$, which results in $q^2 = 2s^2$. Note that q^2 is then even, so q must be even.

But if both p and q are even, this contradicts the assumption that $\gcd(p, q) = 1$: we have found a contradiction.

Thus we can conclude that there do not exist $p, q \in \mathbb{Z}$, with $q \neq 0$ and $\gcd(p, q) = 1$, such that $\sqrt{2} = p/q$. Therefore, $\sqrt{2}$ is irrational. □

Proof by Counter-Example

- Proofs by counter-example are used to show that statements are false.
- General form:
  1. Express the statement to be proved in the form:

     $$\forall x \in D.P(x)$$

     This step is sometimes done mentally.
  2. Find an $x \in D$ such that $P(x)$ is false.

Proof by Counter-Example

Example 9: Let $p(n) = n^2 + n + 41$. Prove that the statement "$\forall n \in \mathbb{N}.p(n)$ is prime" is false.

Proof. Take the counter-example $n = 40$. In this case, we have $p(n) = 1681 = 41 \cdot 41$, which is not prime.

Thus, the statement is false. □

Example 10: Show that the statement "Every positive integer can be written as the sum of the squares of two integers" is false.

Proof. Take the counter-example 3, which is an integer that cannot be written as the sum of the squares of two integers.

To see this, note that the only squares less than 3 are 0 and 1, and the possible sums of these squares are $0 + 0 = 0$, $0 + 1 = 1$, and $1 + 1 = 2$, none of which equal 3.

Thus 3 is a counter-example, and the statement is false. □

Proof by Case Analysis

- Generally used to prove $P \rightarrow Q$.
- The proof divides P into cases and shows that Q follows from each possible case.
- General form:
  1. To show that $P \rightarrow Q$, first show that

$$P \equiv P_1 \vee P_2 \vee \ldots \vee P_n$$

  2. Then, separately, show each of the implications

$$\begin{aligned} P_1 &\rightarrow Q \\ P_2 &\rightarrow Q \\ \ldots &\rightarrow \ldots \\ P_n &\rightarrow Q \end{aligned}$$

Proof by Case Analysis

Example 11: Show that, given two real numbers x and y, $min(x, y) + max(x, y) = x + y$.

Proof. There are only three possibilities for x and y:

$$x < y \quad \text{or} \quad x = y \quad \text{or} \quad x > y$$

.

Let's analyse each case separately:

- If $x < y$, then $min(x, y) + max(x, y) = x + y$.
- If $x = y$, then $min(x, y) + max(x, y) = x + y$.
- If $x > y$, then $min(x, y) + max(x, y) = y + x = x + y$.

Thus, we can conclude that we always have $min(x, y) + max(x, y) = x + y$. □

Proof by Case Analysis

Example 12: Show that $|xy| = |x||y|$, where x and y are real numbers.

(**Note:** $|a|$ is the absolute value of the real number a, defined as $|a| = a$ when $a \geq 0$, and $|a| = -a$ when $a < 0$.)

Proof. Note that there are four possible cases for the combination of x and y:

- x and y are both non-negative,
- x is non-negative and y is negative,
- x is negative and y is non-negative, or
- x and y are both negative.

Proof by Case Analysis

Let's analyse each case separately:

- If x and y are both non-negative, then $xy \geq 0$ and we have
$$|xy| = xy = |x||y|.$$
- If x is non-negative and y is negative, then $xy < 0$ and we have
$$|xy| = -xy = x(-y) = |x||y|.$$
- If x is negative and y is non-negative, then $xy < 0$ and we have
$$|xy| = -xy = (-x)y = |x||y|.$$
- If x and y are both negative, then $xy \geq 0$ and we have
$$|xy| = xy = (-x)(-y) = |x||y|.$$

Thus, we can conclude that the statement is always true. □

Proof of Equivalences

- It is very common to need to show that a set of statements are all equivalent.
- General form:
  1. To show that $P_1 \leftrightarrow P_2 \leftrightarrow ... \leftrightarrow P_n$, separately show each of the implications

$$P_1 \rightarrow P_2$$
$$P_2 \rightarrow P_3$$
$$... \rightarrow ...$$
$$P_n \rightarrow P_1$$

- Important: The proof is not complete if the cycle of implications is not closed, showing that the last proposition implies the first: $P_n \rightarrow P_1$.
- Special case: To prove that $P_1 \leftrightarrow P_2$, one can separately show that $P_1 \rightarrow P_2$ and $P_2 \rightarrow P_1$.

Existence Proof

- A proof of a theorem of the type $\exists x.P(x)$ is called an **existence proof**.
- There are two ways to produce an existential proof:
  1. A **constructive** proof produces an element a such that $P(a)$ is true.

     The element a is called a **witness** of the proof.
  2. A **non-constructive** proof does not produce a witness, but proves $\exists x.P(x)$ in some other way.

     One way is, for example, to produce a proof by contradiction.

Existence Proof: Constructive

Example 13: Show that there exists a positive integer that can be written as the sum of cubes of positive integers in two distinct ways.

Proof. After a laborious search (for example, using a computer program), we find that

$$1\,729 = 10^3 + 9^3 = 12^3 + 1^3.$$

□

The proof above is constructive because it produces a witness (the number 1 729 along with its decompositions) that attests the desired existence.

Existence Proof: Non-Constructive

Example 14: There exist irrational numbers x and y such that x^y is rational.

Proof. We know that $\sqrt{2}$ is irrational (already proven). Consider the number $\sqrt{2}^{\sqrt{2}}$. There are two possibilities for this number:

1. It is rational. In this case, we have two irrational numbers $x = \sqrt{2}$ and $y = \sqrt{2}$ such that x^y is rational.
2. It is irrational. In this case, we can calculate that

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2,$$

which is a rational number. Thus, we have two irrational numbers $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ such that x^y is rational. □

The proof above is non-constructive because it does not produce a witness that attests the desired existence. $y = \sqrt{2}$ satisfy the property, but we do not know which of these two pairs is the correct one!

Common Errors in Proofs

- Arguing from examples.
- **Theorem:** "If $m+n$ is even then $m-n$ is even."

  Incorrect Proof: If $m=14$ and $n=6$ then $m+n=20$ which is even and $m-n=8$ which is also even. □
- Jumping to a conclusion; or claiming something is true without giving a proper reason.
- **Theorem:** "If $m+n$ is even then $m-n$ is even."

  Incorrect Proof: Suppose m and n are integers and $m+n$ is even. By the definition of even, $m+n=2k$ for some integer k. Then $m=2k-n$ and thus $m-n$ is even. □
- Exercise: Correct the proofs above by properly proving the statement "If $m+n$ is even then $m-n$ is even".
- Many of the fallacies we saw in the class on logical inference are common errors in proofs.

The Role of Open Problems

- Some conjectures remain open for a long time before their truth or falsity can be established.
- Even if they fail to prove or disprove the conjecture, mathematicians often make many important advances while trying.

Examples:

1. **Fermat's Last Theorem:** There are no positive integers, for any $n > 2$ x, y, z that satisfy the equation

   $$x^n + y^n = z^n$$

   This conjecture remained open from 1621 until 1994, when it was finally solved.

   While attempting to prove it, some mathematicians created the important field of algebraic number theory (though it did not resolve the conjecture by itself).

2. The biggest open problem in computer science is the P **vs.** NP **Problem**:

   "Is every problem whose solution can be quickly verified by a computer also quickly solvable by a computer?"

Number Theory

Euclid's Division Theorem

Theorem 1. Let a and b be positive integers. There exist integers q and r such that

$$a = bq + r \qquad 0 \leq r < b$$

Moreover, q and r are unique.

Proof. Existence: division algorithm.

1. $q = 0$; $r = a$.
2. If $r < b$, stop. In this case, the quotient is q and the remainder r.
3. If $r \geq b$, set $r := r - b$, $q := q + 1$ and return to step 2.

Uniqueness: Let q, q', r, r' be such that

$$a = bq + r, \qquad 0 \leq r < b \tag{1}$$
$$a = bq' + r', \qquad 0 \leq r' < b \tag{2}$$

Subtracting (1) from (2): $r - r' = b(q' - q)$. Since $0 \leq r, r' < b$, we have $0 \leq r - r' < b$, that is $0 \leq b(q' - q) < b$. Thus $q' = q$ and $r' = r$. □

Question

Does the theorem above work if $a, b \in \mathbb{Z}$?

- First, $b \neq 0$.
- But how to divide 3 by -2?
- Simple: quotient -1 and remainder 1. Indeed,

$$3 = -2(-1) + 1$$

In general:

Theorem 1'. Let a and b be integers, $b \neq 0$. There exist unique integers q and r such that

$$a = bq + r \qquad 0 \leq r < |b|$$

Remarks:

- The algorithm always halts: decreasing sequence of positive integers.
- The result corresponds to the output specification (trivially).
- The algorithm is extremely inefficient, especially if $a \gg b$.

Parity and divisibility

Definition 1. An integer a is **odd**, if there exists an integer k, such that $a = 2k + 1$.

Example 15: Show that -3 is odd.

Solution. $-3 = 2(-2) + 1$ □

Definition 2. $b \in \mathbb{Z}$ is **divisible** by $a \in \mathbb{Z}$, if there exists a $c \in \mathbb{Z}$, such that $b = a.c$.
Notation: $a \mid b$.

Example 16: Let a, b be integers and suppose that $a \mid b$. Show that $a \mid b^2 + 2b$.

Solution.

$$\begin{aligned} b^2 + 2b &= (ak)^2 + 2(ak) \\ &= a(k(ak + 2)). \end{aligned}$$

□

Equivalence Relations and Classes

Definition 3. Let S be a set and $\sim$ a relation on elements of S. We say that $\sim$ is an **equivalence relation** if $\forall x, y, z \in S$:

1. **Reflexive** $x \sim x$.
2. **Symmetric** If $x \sim y$ then $y \sim x$.
3. **Transitive** If $x \sim y$ and $y \sim z$ then $x \sim z$.

Definition 4. If $x \in S$ then the **equivalence class** of x is the set of elements of S that are equivalent to x under $\sim$. We denote:

$$\overline{x} = \{y \in S : y \sim x\}.$$

Properties:

- Any element of an equivalence class is a representative of the whole class.
- S is the union of all equivalence classes.
- Two distinct equivalence classes cannot have a common element.

Quotient set = set of equivalence classes of $\sim$ in S.

Modular arithmetic

Definition 5. The integers a, b are **congruent modulo** n if $n \mid (a - b)$. We write $a \equiv b \pmod{n}$.

Examples:

$$10 \equiv 0 \pmod{5} \qquad 23 \equiv 1 \pmod{11}$$

Proposition 1. Congruence modulo n is an equivalence relation.

Proof.

- $a \equiv a \pmod{n}$ (trivial)
- if $a \equiv b \pmod{n}$, then $a - b$ is a multiple of n. But $b - a = -(a - b)$, so $b - a$ is also a multiple of n. Therefore $b \equiv a \pmod{n}$.
- Transitivity: exercise. □

Example 17: Let $a, b, c, d, n \in \mathbb{Z}$ and suppose that $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$. Show that $a + c \equiv b + d \pmod{n}$.

Solution. Exercise.

Arithmetic Modulo n ($\mathbb{Z}_n$)

Definition 6. The set of **integers modulo** n is denoted by $\mathbb{Z}_n$. That is, if $a \in \mathbb{Z}$, then

$$\overline{a} = \{a + kn \mid k \in \mathbb{Z}\}$$

In particular, $\overline{0}$ is the set of multiples of n.

Euclid's division algorithm: given positive integers a and n, $a > n$, there exist integers q and r such that

$$a = n.q + r \qquad 0 \leq r \leq n - 1$$

That is, $a - r \equiv 0 \pmod{n}$ and therefore $a \equiv r \pmod{n}$.

In other words,

$$\mathbb{Z}_n = \{\overline{0}, \overline{1}, \ldots, \overline{n-1}\}$$

This is pretty cool! ☺

Arithmetic Modulo n ($\mathbb{Z}_n$)

Let $\overline{a}$ and $\overline{b}$ be classes in $\mathbb{Z}_n$. Then,

$$\overline{a} + \overline{b} = \overline{a + b}$$

Example 18:

$$5 + 4 \equiv 9 \equiv 1 \pmod 8$$

Hence,

$$\overline{5} + \overline{4} = \overline{9} = \overline{1}$$

The **difference** between two classes is defined analogously.

The formula for **multiplication** of classes $\overline{a}$ and $\overline{b}$ in $\mathbb{Z}_n$ is:

$$\overline{a}.\overline{b} = \overline{a.b}$$

There is no division modulo! – well, only in some special cases...

Modular Arithmetic

Properties of addition:

A1 $(\overline{a} + \overline{b}) + \overline{c} = \overline{a} + (\overline{b} + \overline{c})$.

A2 $\overline{a} + \overline{b} = \overline{b} + \overline{a}$.

A3 $\overline{a} + \overline{0} = \overline{a}$.

A4 $\overline{a} + \overline{-a} = \overline{0}$.

Modular Arithmetic

Properties of multiplication:

M1 $(\overline{a}.\overline{b}).\overline{c} = \overline{a}.(\overline{b}.\overline{c})$.

M2 $\overline{a}.\overline{b} = \overline{b}.\overline{a}$.

M3 $\overline{a}.\overline{1} = \overline{a}$.

AM $\overline{a}.(\overline{b} + \overline{c}) = \overline{a}.\overline{b} + \overline{a}.\overline{c}$.

Example 19: in $\mathbb{Z}_6$,

$$\overline{2}.\overline{3} = \overline{6} = \overline{0}!!!$$

Oops! Is that a problem??

No, it is not ☺

Divisibility Criteria – Divisibility by 3

Theorem 2. $3 \mid a$ if **the sum of all digits of** a is divisible by 3.

Proof. Let

$$\begin{aligned} a &= a_n a_{n-1} \dots a_1 a_0 \\ &= a_n.10^n + a_{n-1}.10^{n-1} + \dots + a_1.10 + a_0 \end{aligned}$$

Since $10 \equiv 1 \pmod 3$, then $10^k \equiv 1 \pmod 3$ (why?)

Hence

$$a \equiv a_n + a_{n-1} + \dots + a_1 + a_0 \pmod 3$$

Therefore, $a \equiv 0 \pmod 3$ if and only if

$$a_n + a_{n-1} + \dots + a_1 + a_0 \equiv 0 \pmod 3$$

□

Divisibility by 9 works the same way!

Divisibility by 11

Theorem 3. $11 \mid a$ if the **alternating sum of all digits of** a is divisible by 11.

Proof. Note that $10 \equiv -1 \pmod{11}$. Therefore,

$$10^k \equiv (-1)^k \pmod{11}$$

is equal to 1 or -1 depending on the parity of k.

Hence,

$$a \equiv (-1)^n . a_n + (-1)^{n-1} . a_{n-1} + \ldots + a_2 - a_1 + a_0 \pmod{11}$$

□

Powers

Example 20: Compute $35^{15} \pmod{20}$.

Solution. Write the exponent in base 2: $15 = 2^3 + 2^2 + 2 + 1$.

Then,

$$35^{15} = 35^{2^3+2^2+2+1} = 35 \cdot 35^2 \cdot (35^2)^2 \cdot ((35^2)^2)^2.$$

Since $35 \equiv 15 \pmod{20}$ and $15^2 \equiv 5 \pmod{20}$, we get

$$\begin{aligned} 35^{15} &\equiv 15 \cdot 5 \cdot 5 \cdot 5 \pmod{20} \\ &\equiv 15 \cdot 5 \pmod{20} \\ &\equiv 15 \pmod{20}. \end{aligned}$$

This shows that modular arithmetic allows us to simplify large powers by reducing intermediate results modulo n.

Prime numbers

Prime and composite numbers

$p \in \mathbb{N}$ is prime if $p \neq 1$ and the only divisors of p are p and 1. If $n \in \mathbb{N}, n \neq 1$ is not prime, then it is composite.

Example 21: Show that 2 is prime.

Solution. Clearly $2 \leq 2$. Let m be a factor of 2. Since 2 is positive, by the size bounds on factors, we have that $m \leq 2$ and $1 \leq m$. The only natural numbers m satisfying these restrictions are 1 and 2. □

Euclidean Algorithm

Definition 7. Let $a, b, d \in \mathbb{Z}$. d is the **greatest common divisor**, notation $d = \gcd(a, b)$, if

- $d \mid a$
- $d \mid b$
- if d' divides both a and b, then $d' \mid d$

If $\gcd(a, b) = 1$ then a and b are relatively prime.

Algorithm for calculating the $\gcd(a, b)$:

- Given $a \geq b > 0$, divide a by b, obtaining remainder r_1.
- If $r_1 \neq 0$, divide b by r_1, obtaining remainder r_2.
- If $r_2 \neq 0$, divide r_1 by r_2, and so on.

The last nonzero remainder in this sequence of divisions is $\gcd(a, b)$.

Euclidean Algorithm

Example 22:

$$\begin{aligned} 1234 &= 54 \cdot 22 + 46 \\ 54 &= 46 \cdot 1 + 8 \\ 46 &= 8 \cdot 5 + 6 \\ 8 &= 6 \cdot 1 + 2 \\ 6 &= 2 \cdot 3 + 0 \end{aligned}$$

Therefore, $\gcd(1234, 54) = 2$.

Questions:

1. Why is the last nonzero remainder the gcd?
2. Why does the algorithm always stop?

Answers:

- Second question: finite sequence: $b > r_1 > r_2 > \cdots \geq 0$. So the algorithm always halts with number of divisions is at most b (why?).
- First question: proof of the Euclidean algorithm.

Euclidean Algorithm

Lemma 2. Let a, b be positive integers. If there exist integers g, s such that $a = bg + s$, then $\gcd(a, b) = \gcd(b, s)$.

Proof. Let $d_1 = \gcd(a, b)$ and $d_2 = \gcd(b, s)$. There exist integers u, v such that $a = d_1 u,\ b = d_1 v$.

Substituting into $a = bg + s$ gives

$$s = d_1 u - d_1 vg = d_1(u - vg).$$

Thus d_1 is a common divisor of b and s.

But d_2 is the greatest common divisor of b and s, hence $d_1 \leq d_2$.

A symmetric argument gives $d_2 \leq d_1$. Therefore, $d_1 = d_2$. □

Euclidean Algorithm

Theorem 4.[Correctness of the Euclidean Algorithm] Let $a, b > 0$. The last nonzero remainder of the Euclidean algorithm applied to a and b is the greatest common divisor of a and b.

Proof. Consider the sequence of divisions:

$$\begin{aligned}
a &= bq_1 + r_1, & 0 \le r_1 < b \\
b &= r_1 q_2 + r_2, & 0 \le r_2 < r_1 \\
r_1 &= r_2 q_3 + r_3, & 0 \le r_3 < r_2 \\
&\vdots & \\
r_{n-2} &= r_{n-1} q_n + r_n, & r_n = 0
\end{aligned}$$

From the last line, r_{n-1} divides r_{n-2}, hence $\gcd(r_{n-1}, r_{n-2}) = r_{n-1}$.

Applying the lemma repeatedly yields $\gcd(a, b) = r_{n-1}$. □

Extended Euclidean Algorithm

Theorem 5. Let $a, b > 0$ and $d = \gcd(a, b)$. There exist integers α, β such that

$$\alpha a + \beta b = d.$$

Example 23: Let $a = 1234$, $b = 54$:

$$\begin{aligned} 1234 &= 54 \cdot 22 + 46 && \Rightarrow 46 = 1234 - 54 \cdot 22 \\ 54 &= 46 \cdot 1 + 8 && \Rightarrow 8 = 54 - 46 \\ & && \Rightarrow 8 = 54 - (1234 - 54 \cdot 22) \\ & && \Rightarrow 8 = 54 \cdot 23 - 1234 \end{aligned}$$

Continuing:

$$\begin{aligned} 46 &= 8 \cdot 5 + 6 && \Rightarrow 6 = 46 - 8 \cdot 5 \\ & && \Rightarrow 6 = (1234 - 54 \cdot 22) - (54 \cdot 23 - 1234) \cdot 5 \\ & && \Rightarrow 6 = 1234 \cdot 6 - 54 \cdot 137 \\ 8 &= 6 \cdot 1 + 2 && \Rightarrow 2 = 8 - 6 \\ & && \Rightarrow 2 = (54 \cdot 23 - 1234) - (1234 \cdot 6 - 54 \cdot 137) \\ & && \Rightarrow 2 = -1234 \cdot 7 + 54 \cdot 160 \end{aligned}$$

Fundamental Lemma

Lemma 3. Let a, b, c be positive integers and suppose that $gcd(a, b) = 1$. Then:

1. If b divides the product $a.c$ then b divides c.
2. If a and b divide t then the product $a.b$ divides t.

Proof.

1. $gcd(a, b) = 1$. By the Extended Euclidean Algorithm, there exist α and β such that

$$\alpha.a + \beta.b = 1$$

Then,

$$\alpha.a.c + \beta.b.c = c$$

Since b divides $a.c$ by the hypothesis (1) and since b divides $\beta.b.c$, then b divides c.

2. If a divides t, then $t = ac$. But b also divides t. Since $gcd(a, b) = 1$, by statement (1), b divides c. Therefore, $c = b.k$ for some integer k and hence,

$$t = a.c = a.b.k$$

□

Fundamental Property of Primes

We can use the fundamental lemma to prove the following property:

Fundamental Property of Primes. Let p be a prime and a and b positive integers. If p divides the product $a.b$, then p divides a **or** p divides b.

The proof is left as an exercise (do it!).

Fundamental Theorem of Arithmetic – Existence

Theorem 6. Given a positive integer $n \geq 2$, we can always write it uniquely as:

$$n = p_1^{e_1} . \ldots . p_k^{e_k}$$

where $p_1 < p_2 < \ldots < p_k$ are prime numbers and $e_1, \ldots, e_k$ are positive integers (multiplicities).

Proof. Naive algorithm: given a positive integer $n \geq 2$, try to divide n by each integer from 2 to $n - 1$.

If one of these integers (say k) divides n, then we find a prime factor of n.

If not, n is prime. Repeat the process for n/k until reaching 1.

Notice that k is prime.

Indeed, if k were composite $\Rightarrow k = a.b$ with $1 < a, b < k$.

But $k \mid n$, so there exists an integer c such that $n = k.c$. Hence,

$$n = a.b.c \quad \text{ABSURD!}$$

Question: When should we stop the search? At $n - 1$?

Fundamental Theorem of Arithmetic – Uniqueness

Let n be the smallest positive integer that admits two distinct factorizations.

We can write:

$$n = p_1^{e_1} \cdot \ldots \cdot p_k^{e_k} = q_1^{r_1} \cdot \ldots \cdot q_s^{r_s}$$

where $p_1 < p_2 < \ldots < p_k$ and $q_1 < q_2 < \ldots < q_s$ are primes and $e_1, \ldots, e_k, r_1, \ldots, r_s$ are positive integers.

Since p_1 divides n, by the **fundamental property of primes** p_1 must divide one of the factors on the right-hand side product.

But a prime can only divide another if they are equal. So $p_1 = q_j$ for some j between 1 and s. Thus,

$$\begin{aligned} n = p_1^{e_1} \cdot \ldots \cdot p_k^{e_k} &= q_1^{r_1} \cdot \ldots \cdot q_j^{r_j} \cdot \ldots \cdot q_s^{r_s} \\ &= q_1^{r_1} \cdot \ldots \cdot p_1^{r_j} \cdot \ldots \cdot q_s^{r_s} \end{aligned}$$

We can then cancel p_1 which appears on both sides of the equation, obtaining

$$m = p_1^{e_1 - 1} \cdot \ldots \cdot p_k^{e_k} = q_1^{r_1} \cdot \ldots \cdot p_1^{r_j - 1} \cdot \ldots \cdot q_s^{r_s}$$

where m is a number smaller than n that has two distinct factorizations.

ABSURD since this contradicts the minimality of n. □

Sieve of Eratosthenes

The sieve of Eratosthenes is the oldest method for finding primes.

Step 1: List the odd numbers from 3 to n.

Step 2: Find the first number k in the list. Cross out the other numbers in the list, k by k.

Step 3: Repeat step 2 until reaching n.

Observations:

1. We can stop at $\sqrt{n}$...
2. We can start crossing out from k^2...

Revised Sieve of Eratosthenes

Step 1: Create a vector v with $\frac{n-1}{2}$ entries, all initialized with the value 1; set $P = 3$.

Step 2: If $P^2 > n$, output the numbers $2j + 1$ for which the j-th entry of v is 1, and stop.

Step 3: If the entry at position $\frac{(P-1)}{2}$ of v is 0, increment P by 2 and return to Step 2.

Step 4: Set $T = P^2$; set the entry at position $\frac{(T-1)}{2}$ to 0 and increment T by $2P$; repeat until $T > n$; then increment P by 2 and return to Step 2.